Topic 1:Diagnosing and Remedying Nonlinearity in Bike-Sharing Demand Models Proposal

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1. Research Question

For a bike-sharing platform to be financially sustainable, it must be able to forecast its prospective demand. We aim to study if we can build a model that takes in weather, time, and seasonal variables to predict demand.

Specifically, does the relationship between weather conditions and temporal factors and hourly bike rental demand follow a linear form, or is there substantive nonlinearity that requires transformation?

2. Data

Source	UCI Machine Learning Repository
Name of Data Set	Bike Sharing (Hourly)
Link	https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset
Period	January 2011 - December 2012
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0) license.
Unit of Observation	Count of one hour of bike rental activity at a specific time
Sample Size	17,379
Variables	17 columns (10 features + 3 control + 3 target + 1 ID) : instant, dteday, season, yr, mnth ,hr, holiday, weekday, workingday, weathersit, temp, atemp, hum, windspeed, casual, registered, cnt (data dictionary can be found in the link above)
Processing Steps	cnt is set aside as the target. instant (row index), dteday (redundant with yr and mnth), and casual and registered (which sum exactly to cnt are dropped. atemp is excluded for near-perfect collinearity with temp. weekday is not used because its information is largely captured by workingday and holiday. No missing values remain, so no imputation is needed.

3. Plan

a. Methods, models and the choices we expect to make

First, we test if a multiple linear regression model on the original variables holds predictive power.

$$cnt = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_j x_j + \varepsilon$$

$$where\ x = \{season,\ yr,\ mnth,\ hr,\ holiday,\ workingday,\ weathersit,\ temp,\ hum,\ windspeed\}$$

$$H_0: \beta_1 = \beta_2 = \ldots = \beta_j = 0 \qquad H_1: at\ least\ one\ \beta_j \neq 0$$

We evaluate the overall F-test for significance to understand if the base model holds some predictive power. We will note the explanatory power (R^2, Adjusted R^2) and residual standard error (RSE) as metrics for communication, and evaluate the residual plots vs fitted, Q-Q, externally studentised, Cook's Distance, Leverage and DFFITS for further analysis of non-linearity and outliers. Partial F-tests (ANOVA) will justify variable inclusion by testing if omitting specific predictors, like windspeed, significantly reduces explanatory power.

b. Potential issues we expect in the data and how we plan to resolve them

1) Nonlinear relationships between predictors and bike demand

Using a baseline linear regression model assumes that a one unit change in a predictor would lead to a constant change in expected demand, holding other variables constant. However, this may not be appropriate for hours of a day as demand does not always increase or decrease. Temperature may also have a nonlinear association with demand, with increases becoming smaller or even decreasing at high temperatures.

To check for the potential nonlinear relationships, we will examine the baseline's residuals-versus-fitted plot. We will also examine the added-variable plots for hour, temperature and weather conditions (humidity, windspeed and weathersit) to investigate each predictor's relationship with demand after controlling for other predictors.

We will test if polynomial factors improve the model, formally a likelihood ratio test:

$$cnt = \beta_0 + \beta_1 x_1 + \ldots + \beta_j x_j + \theta_1 hr^2 + \theta_2 temp^2 + \ldots + \varepsilon$$

$$H_0: \theta_1 = \theta_2 = \ldots = \theta_k = 0 \qquad H_1: at\ least\ one\ \theta_k \neq 0$$

We will also consider if dummy variables are better suited for this problem.

2) Heteroskedasticity, unequal residual variance

Linear regression assumes that errors have constant variance. However, high-demand periods may have larger fluctuations than quiet periods and residual variability may differ between working and non-working days. This could make conventional standard errors and prediction intervals unreliable.

We will examine the residual-versus-fitted and scale-location plots. A funnel-shaped spread in the residual plot or upward-sloping pattern in the scale-location plot would suggest that residual variance increases with fitted demand. A roughly flat band in the scale-location plot would be more consistent with constant variance.

If the diagnostics and tests indicate that residual variance increases with fitted demand, we will test a log transformation of cnt using a Box-Cox procedure of the form $cnt^{(\lambda)}$ where $\lambda = 0 \Rightarrow log(cnt)$.

$$H_0: \lambda = 0 \qquad H_1: \lambda \neq 0$$

c. Potential extensions

We will look at regularisation terms, such as Lasso and Ridge regression to reduce overfitting and improve generalisation. We will also evaluate using an expanding window validation method, as this is a time series dataset. Lastly, we will look at principal component analysis as a tool in the Linear Regression Pipeline.